E41: the order-24 walk's step shortfall on E18's $j = 0$ curves — sampling noise, not a walk defect; the corrected reference

Run report and results, 2026-11-02. Note §9.52 (pre-registration, pushed `f99a56359e` before any fix code or registered run) and §9.53 (results). Stage 0 code `089f7f7823` / `6115e0ea0a`; fix and harness `b0e307d9b8` (`src/cryptanalysis/twist24_shortfall.rs`, option `Jumps`, default `Jumps::E34`; `examples/e41_run.rs`), tables `examples/e41_tables.rs` at `5d15482969`. Binaries built from clean `git archive` exports (blob-checked, 0 mismatches), sha256 in `experiments/48_twist24_shortfall/binary_sha256.txt`. Records `experiments/48_twist24_shortfall/` (the three records over 1 MiB are kept by sha256; the store push needs the `aws` CLI, absent here). Host: the shared 4-vCPU container, one core per process, isolab-eks unreachable; **operation counts only**, no wall-time claim. Python only lays out recorded values (`figures/plot.py`); every number comes from the Rust examples.

Question

E34 (§9.39) measured the cheap order-24 $\langle \pm\psi \rangle$ walk at 1.701 ± 0.111 times fewer steps than the order-6 walk (16 walks per curve), against the $\sqrt{24/6} = 2$ of ideal walks; its 1 GAE lead over the cheap order-6 walk (1.026) was within noise. Where does the shortfall come from, and does a fix make the walk measurably cheaper across the §5a band? The walk is the matched rho reference: a cheaper walk moves the reference, never an index-calculus cost.

Stage 0: the shortfall by cause (counts only, before the registration)

E34's stored walks put the order-6 walk at 0.924 ± 0.046 of its ideal $\sqrt{\pi(n-1)/2m}$ and the order-24 walk at 1.086 ± 0.045 (128 walks each): the 1.701 is the two arms' sampling errors in opposite directions. On 512 fresh walks each (seeds 41000–41063) the instrument finds order 6 at 1.027 ± 0.023 and order 24 at 1.038 ± 0.023. By cause: distinguished-point losses 0 (exhaustive storage); fruitless cycles $0.84 \cdot 10^{-4}$ revisits a step for order 24 (98% 2-cycles), costing $< 10^{-4}$ of the ideal; look-ahead redirects $1/r$ in both orders; representative ties ($x_0 = 0$ inputs) cost no step. No order-24-specific cause above 0.1%. The only non-noise component is a small excess common to both orders, consistent with the r-adding factor $1/\sqrt{1 - 1/r}$.

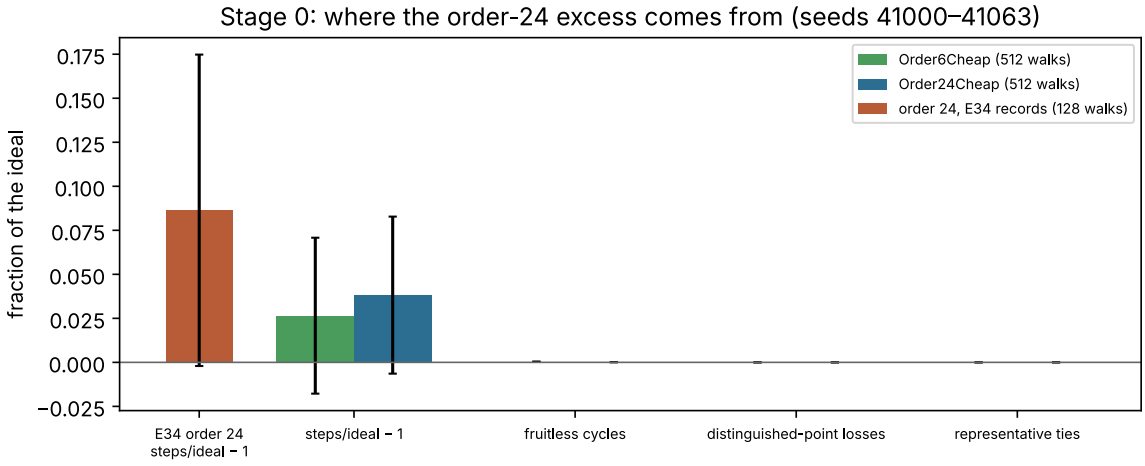


Figure 1: Stage 0: steps/ideal − 1 and the share of each cause, 512 fresh walks per order (bars 95 %), with E34's stored order-24 walks for comparison. Source: `stage0_run.json`, `stage0_records.json`.

Fix and arms

Fix F (chosen from Stage 0, null prediction): E34's 32 jumps plus $R_{32+j} = R_j + R_{(j+1) \bmod 32}$, $r = 64$; added work 32 group additions and 64 mod-n operations a walk, charged. Arms on E18's eight curves: O6B (E29's order 6), O6C (cheap order 6), O24 (E34's cheap order 24), O24L (its mul-lean twin), O24F (O24 with F), and the A/A arm O24′ (O24 on a disjoint seed stream). 128 walks per curve and arm, schoolbook (seeds 48000+k; A/A 49000+k), 32 per curve in Karatsuba. Band: F_p multiplications (set-up included, $[d]G$ excluded) $+w \cdot 97 \cdot$ native operations, $w \in \{0, 0.1, 1\}$, pooled $\sum \frac{\text{cost}}{\sqrt{n}}$. Resolved = 95 % interval excludes 1 at all three weights and $|\text{ratio} - 1|$ exceeds the A/A deviation.

Checks

- **Identity (fix off)**: 1,280 of 1,280 E34 Stage R records recomputed byte-identical (serialized); E27's walk code (`quartic_folds.rs`) is untouched by every E41 commit.
- **Correctness**: 0 wrong logs in 7,424 walks (6,144 schoolbook, 1,280 Karatsuba).

- **Positive control** (F with b_{j+1} dropped): wrong log flagged on 32 of 32 walks.
- **Added cost of F**: exactly 32 additions and 64 mod-n operations on all 1,280 F walks; zero on every other arm. Redirects fall to $1/64$, fruitless revisits $3.6\times$.
- rustfmt (`--edition 2021`) and clippy (`-D warnings`) clean on E41′s files.

Steps against the ideal (schoolbook, 1,024 walks per arm)

O6B 0.969 ± 0.016; O24 1.006 ± 0.016; O24F 0.999 ± 0.016; O24′ 0.987 ± 0.016. Steps ratio order 6 / order 24 pooled per $\sqrt{n}$: 1.927 (main stream), 1.965 (against the A/A stream); E34 1.701.

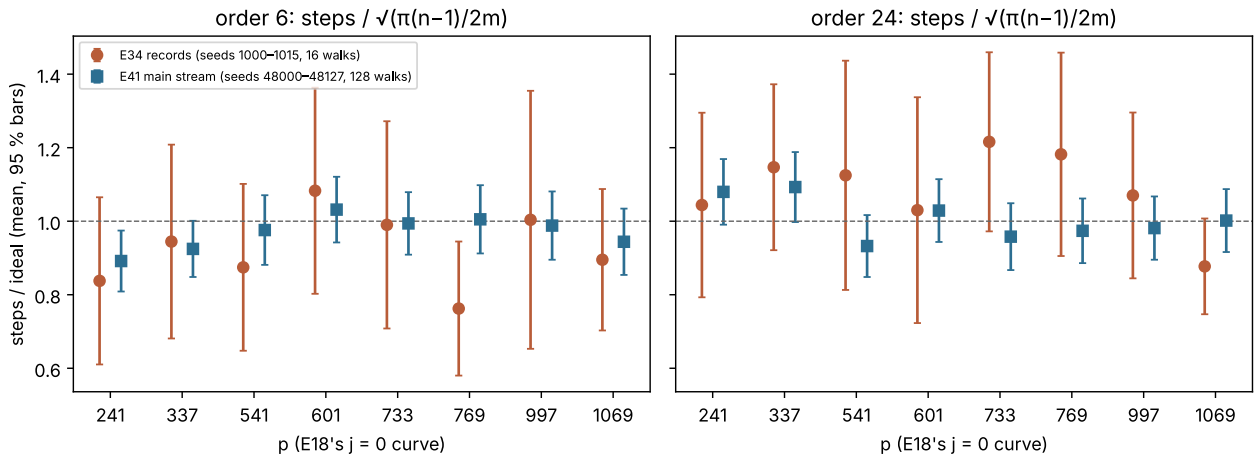


Figure 2: Steps over the ideal per curve: E34′s 16 stored walks (orange) and E41′s 128 fresh walks (blue), bars 95 %. Source: `stage0_records.json`, `summary.json`.

The band (pooled; > 1: the second arm cheaper; 95 % intervals)

ratio	arith.	0 GAE	0.1 GAE	1 GAE	resolved
O6B / O24	school	1.761 [1.686, 1.836]	1.418 [1.356, 1.481]	1.275 [1.218, 1.332]	yes (8/8 curves)
O6C / O24	school	1.696 [1.623, 1.768]	1.320 [1.262, 1.378]	1.162 [1.110, 1.214]	yes (7/8; 1 unresolved)
O24 / O24F	school	1.021 [0.978, 1.064]	1.019 [0.975, 1.064]	1.018 [0.973, 1.064]	no
O24 / O24′ (A/A)	school	1.019	1.019	1.020	floor
O6B / O24	Karatsuba	1.711	1.369	1.249	yes
O6C / O24	Karatsuba	1.635	1.267	1.137	yes
O24 / O24F	Karatsuba	0.984	0.981	0.979	no

E34 (16 walks): O6B / cheap 1.563, 1.254, 1.126; O6C / cheap 1.505, 1.167, 1.026.

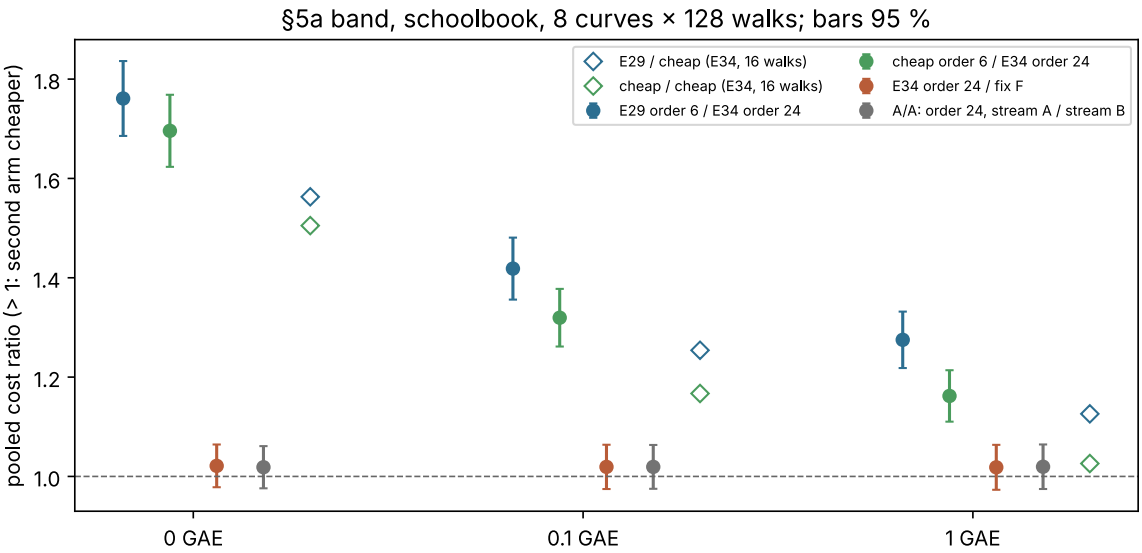


Figure 3: Pooled cost ratios across the §5a band with 95 % intervals, schoolbook, 8 curves × 128 walks; hollow diamonds are E34′s 16-walk values. Source: `summary.json`.

Verdicts (§9.52)

G2.216, G2.217, G2.218 (kill gates) hold. **G2.219 falsified**: O6B ran at 0.969 of its ideal, under the registered $[0.98, 1.08]$; O24 1.006 and the steps ratio 1.927 are inside. G2.220 holds (A/A 1.019–1.020). G2.221 holds (F 0.75% fewer steps; theory 0.8%). G2.222 holds: F is not measurably cheaper (null). G2.223 and G2.224 hold: the corrected reference is resolved at every weight against both order-6 walks, including 1.162 at 1 GAE against the cheap order-6 walk (E34: 1.026, within noise). G2.225 (7/8 curves) and G2.226 (Karatsuba, same orders) hold.

Consequences

E18′s ψ rows (reference correction). Against E29′s order-6 walk, the band-ranked order-24 walk is 1.761 ×, 1.418 × and 1.275 × cheaper at 0, 0.1 and 1 GAE (E34: 1.563, 1.254, 1.126). E18′s ψ-arm distances to rho lengthen by these factors, and the modelled handover moves later by 3.27, 2.02 and 1.40 bits (E34: 2.58, 1.31, 0.68). The unranked point estimates for the cheapest arm at each weight are 1.838 (O24L), 1.446 and 1.298 (O24F). **IC rows (E31 solver A/B with E27 products, 20 rows on the GLS curves).** Order 24 does not apply there ($\mathrm{Aut} = \{\pm 1\}$), and E41 changes no walk on those curves, so their reference stays walk (iii) and every distance is unchanged (factor 1). Both columns are listed in `tables.md`: for example, $p = 101$ seed 1 arm A gives $7.664 \cdot 10^4$, $2.672 \cdot 10^5$, $1.371 \cdot 10^6$ before and after. Rho is ahead of every row at every weight.

What it says

E34′s step shortfall was sampling noise in a 16-seed design. Both walks sit within a few per cent of their own $\sqrt{|G|}$ ceiling, and no order-24-specific defect exists to fix. F, which targets the r-adding excess common to both walks, is not measurably cheaper. The matched rho on E18′s $j = 0$ curves is stronger than E34 reported, and resolved at every weight. **Class: a reference correction**, not an index-calculus improvement: E18′s rows move further from rho, and the GLS IC rows do not move. No leaderboard, `benchmarks/` or validation-gate change is made here.

Limitations

One host, no isolation tier, operation counts only. Steps are random: the per-arm SE is 1.6% at 1,024 walks, so effects below about 3% (F's) are not resolvable. Curves run from 32 to 40 bits, and the set-up share shrinks with n. The records over 1 MiB are not in the store (no `aws` CLI); they are identified by sha256. Unit tests were not run in this session (release unit-test builds are excluded here); the identity run covers the instrument's and the option's equality with E34.